

Calculus – Group Discussion

Week 7 – Multivariable optimization

ADSAI & CS / PR12

Learning goals (Week 7)

- Determine critical points of functions of two variables.
- Classify critical points using the second derivative test.
- Solve constrained optimisation problems using Lagrange multipliers.

Critical points for multivariable functions

For a differentiable function $f(x, y)$, a point (a, b) is a **critical point** if

$$\frac{\partial}{\partial x} f(a, b) = 0 \quad \text{and} \quad \frac{\partial}{\partial y} f(a, b) = 0.$$

Equivalently,

$$\nabla f(a, b) = \mathbf{0}.$$

Geometric meaning. When $\nabla f(a, b) = \mathbf{0}$, the surface has no first-order change in either coordinate direction. So the tangent plane is horizontal, which implies:

- a local minimum,
- a local maximum,
- or a saddle point.

Second derivative test for multivariable functions

For a twice differentiable function $f(x, y)$, the matrix

$$H_f(x, y) = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix}$$

is called the **Hessian matrix**. The sign of the determinant

$$\det(H_f) = D(x, y) = f_{xx}(x, y)f_{yy}(x, y) - (f_{xy}(x, y))^2.$$

tells us about its **definiteness**. At a critical point (a, b) :

Positive definite: If $D(a, b) > 0$ and $f_{xx}(a, b) > 0$, then f has a **local minimum**.

Negative definite: If $D(a, b) > 0$ and $f_{xx}(a, b) < 0$, then f has a **local maximum**.

Indefinite: If $D(a, b) < 0$, then (a, b) is a **saddle point**.

Inconclusive: If $D(a, b) = 0$, the test is inconclusive.
 f could have a local max/min, or saddle point at (a, b) .

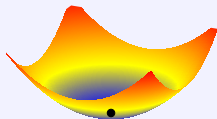
Geometry of the second derivative test

Local minimum

$$D(a,b) > 0$$

$$f_{xx}(a,b) > 0$$

minimum

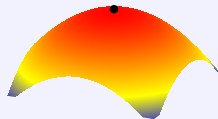


Local maximum

$$D(a,b) > 0$$

$$f_{xx}(a,b) < 0$$

maximum



Saddle point

$$D(a,b) < 0$$

$f_{xx}(a,b)$ not decisive

saddle



Visualization: <https://www.geogebra.org/m/q5edhjpc>

Lagrange multipliers for constrained optimization

Suppose we want to optimize $f(x, y)$ subject to the constraint

$$g(x, y) = c.$$

Geometrically, we are searching for the highest or lowest value of f *along the constraint curve*.

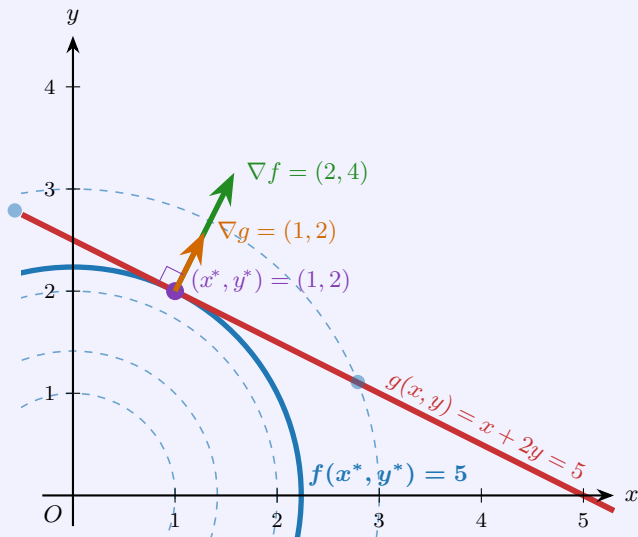
At an optimal point on the constraint, the level curve of f is tangent to the constraint curve. That means their normals are parallel:

$$\nabla f(x, y) = \lambda \nabla g(x, y).$$

Together with the constraint

$$g(x, y) = c,$$

this gives a system of equations to solve.



Problem

Minimize $f(x, y) = x^2 + y^2$
 subject to $g(x, y) = x + 2y = 5$

Lagrange Condition

$$\nabla f(x, y) = \lambda \nabla g(x, y)$$

$$(2x, 2y) = \lambda(1, 2)$$

$$\hookrightarrow y = 2x$$

Plug back into constraint

$$x + 2(2x) = 5 \implies x^* = 1$$

$$y^* = 2x^* = 2$$

$$\therefore (x^*, y^*) = (1, 2).$$

Visualization: <https://www.desmos.com/calculator/xmzby9u8nm>

Key takeaways

- Critical points of $f(x, y)$ occur where both first partial derivatives vanish.
- The Hessian matrix and the discriminant $D = f_{xx}f_{yy} - f_{xy}^2$ classify local behavior.
- Positive definite behavior gives a local minimum; negative definite gives a local maximum; indefinite gives a saddle.
- Constrained optimization restricts the search to a curve or surface.
- Lagrange multipliers work because, at an optimum on a constraint, the gradients are parallel

$$\nabla f = \lambda \nabla g$$